



Dr. Madhu Pahar

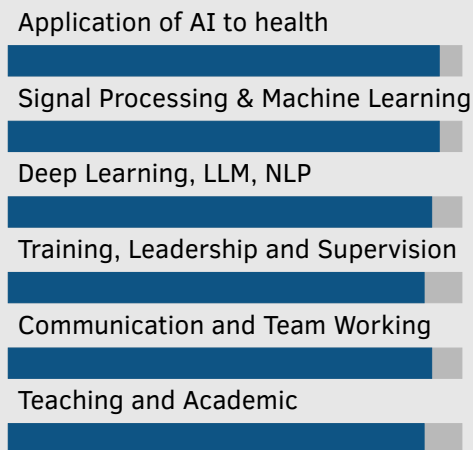
Senior Research Fellow,
University of Sheffield, UK

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About me

Madhu is an experienced ML and AI engineer & researcher, specialising in multimodal analysis of speech, audio, and biological data for real-world clinical applications. Previously, he led the development of AI systems for detecting TB and COVID-19 from respiratory audio and sensor data in clinical environments in South Africa, and developed deep learning methods for genomic and transcriptomic analysis using Oxford Nanopore sequencing data in the USA. Currently, he is working on multimodal speech and video analysis to detect early signs of cognitive decline, including mild cognitive impairment and dementia, using large-scale NHS datasets. His research focuses on developing interpretable, robust, and deployable ML systems, working closely with clinicians and interdisciplinary teams to translate AI into practical healthcare solutions.

Overall Skills



Education

- 2010-2016 Ph.D. in Computational Neuroscience University of Stirling, Scotland, UK
Thesis title: A Novel Sound Reconstruction Technique based on a Spike Code (event) Representation
- 2009-2010 MSc in Computing for Financial Markets University of Stirling, Scotland, UK
Thesis title: Implementation of VaR and Option Strategies
- 2005-2009 BSc in Mathematics University of Calcutta, INDIA
Mathematics (LPP, C, Probability distribution, Boolean Algebra, Numerical Analysis, Game theory, Differential Equation, Set theory), Statistics (Sampling Distribution, Hypothesis, ANOVA, Design), Physics (Mechanics, Modern Physics, Electronics)

Selected First-Author Research Outputs

- 2026 Pahar, M. *et al.*, 2026. PROCESS-2: A Benchmark Speech Corpus for Early Cognitive Impairment Detection. *arXiv:2605.14888* (Under review: Nature Scientific Data)
- 2025 Pahar, M. *et al.*, 2025. Automatic detection of early cognitive decline using multimodal feature fusion and transfer learning on real-world conversational speech. *IEEE Journal of Biomedical and Health Informatics*, 29(12). pp.8727-8734. IEEE
- 2025 Pahar, M. *et al.*, 2025. CognoSpeak: an automatic, remote assessment of early cognitive decline in real-world conversational speech. In *2025 IEEE Symposium on Computational Intelligence in Health and Medicine (CIHM)*. (pp 1-7). IEEE
- 2024 Pahar, M. and Liu, Q., 2024. NanoCNN: A CNN-Based Tool to Detect RNA m⁶A Methylations Using Oxford Nanopore Sequencing. In *2024 4th International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)* (pp. 1-6). IEEE
- 2023 Pahar, M. *et al.*, 2023. Accelerometer-based Bed Occupancy Detection for Automatic, Non-invasive Long-term Cough Monitoring. *IEEE Access*, vol. 11, pp. 30739-30752. IEEE
- 2022 Pahar, M. *et al.*, 2022. Automatic Tuberculosis detection in cough patterns using NLP-style cough embeddings. In *2022 International Conference on Engineering and Emerging Technologies (ICEET)* (pp. 1-6). IEEE
- 2022 Pahar, M. *et al.*, 2022. COVID-19 detection in cough, breath and speech using deep transfer learning and bottleneck features. *Computers in Biology and Medicine*, 141, p. 105153. Elsevier
- 2022 Pahar, M. *et al.*, 2022. Automatic non-invasive Cough Detection based on Accelerometer and Audio Signals. *Journal of Signal Processing Systems*. Springer
- 2022 Pahar, M. *et al.*, 2022. Wake-Cough: cough spotting and cough identification for personalised long-term cough monitoring. In *2022 30th European Signal Processing Conference (EUSIPCO)* (pp. 185-189). IEEE
- 2021 Pahar, M. *et al.*, 2021, June. Deep Neural Network based Cough Detection using Bed-mounted Accelerometer Measurements. In *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 8002-8006). IEEE
- 2021 Pahar, M. *et al.*, 2021. COVID-19 cough classification using machine learning and global smartphone recordings. *Computers in Biology and Medicine*, 135, p.104572, Elsevier
- 2021 Pahar, M. *et al.*, 2021. Automatic Cough Classification for Tuberculosis Screening in a Real-World Environment. *Physiological Measurement*, 42(10), p.105014. IOPscience

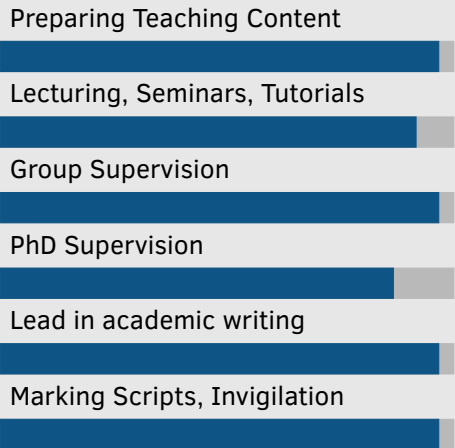
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Core Expertise



Teaching Skills



Academic Output and Interactions

Funding Applications

Successful (senior investigator): EDCTP, CAGE-TB collaborative project between Stellenbosch University (South Africa), Germany, Netherlands and Uganda, €3m
Unsuccessful (co-PI), re-applying: NIHR i4i (invention for innovation), developing and implementing CognoMemory into Memory assessment pathways, collaboration with Sheffield Institute for Translational Neuroscience (SITraN), £1.5m

Editorial Board Member

Computational Biology and Bioinformatics, Journal of Artificial Intelligence in Modern Medicine

Peer Review Team Member

IEEE Journal of Biomedical and Health Informatics, IEEE Transactions on Biomedical Circuits and Systems, Journal of Biomedical Informatics, Biomedical Signal Processing and Control, Computers in Biology and Medicine, ICASSP, Interspeech

Invited talks, Keynote Addresses, Session chairs

2026	EDI Focused Research Showcase, Session Chair at Interspeech 2026
2024 - 2026	CMI Townhall
4 July, 2025	Insigneo Showcase 2025
23 June, 2025	SLT-CDT
13 May, 2024	Pint of Science Artificial brains diagnosing drains (and disease!)
11 May, 2023	2nd Global Webinar on AI, Machine Learning & Data Science
7-9 Oct, 2022	4th International Conference on AI (AIAM2022), IEEE, Germany
29-2 Sep, 2022	Session chair of EUSIPCO 2022 at Belgrade, Serbia
20-22 July, 2022	Session chair of ICECET 2022 at Prague, Czech Republic
18 July, 2022	AI Expo, Belgium
1-3 Dec, 2021	CHPC National Conference 2021, South Africa
10-12 Nov, 2021	Organizer and session chair of 2021 Postdoctoral Conference “Beyond The Pandemic: Research Now And In The Future”, South Africa
28-29 Oct, 2021	National Seminar cum Workshop at BIT, Mesra, India
2020, 2021	CHPC (<i>Advancing high-performance computing in South Africa</i>)
17-18 July, 2020	ELECTROSYMP-20, VIT Bhopal University, India

International Collaborations, Academic Responsibilities & Awards

2025	Best-paper award at JBHI
2024	Successfully established collaborations with the Sheffield Institute for Translational Neuroscience (SITraN) its past members like Professor Annalena Venneri and her group for investigating data cohorts consisting attention, memory and executive functions to detect early signs of cognitive decline and willing to expand the collaboration
2024	Successfully embarked on a collaboration with Dr Chaona Chen to use facial cues collected from the CognoSpeak video interview to analyse dementia, mood disorder and depression and possibly extending it to Dr. Lua’s research group in Vietnam via Mental Health Workshop
2024	Actively teaching COM4520 Darwin Project and co-supervising undergraduate and PhD students and participating in neurodivergent group at the School of Computer Science at the University of Sheffield
2024-2027	Member of Senate Appeals and Discipline Panels at the University of Sheffield (Reference: K Emma Brace)
2023	Co-supervised and supported a PhD student in bioinformatics and genetic data analysis for RNA sequencing at UNLV
2021-2023	Monitoring the long-term recovery of TB patients with the biomedical team at UPPSALA university (Sweden) & TASK (South Africa)
2019-2023	Successful co-supervision of 2 masters and 2 skripsie (short-term undergraduate project) students at Stellenbosch University
2022	Mentored students from University of Exeter, Chinese University of Hong Kong, Stellenbosch University, IE University and University of São Paulo, representing four different continents, in UN’s Future 17 Sustainable Development Goals Programme
2022	Executive committee member of postdoc society (PDS Exec 2022)
2021	Regional judge at Eskom Expo for young scientists
2021	Top-20 postdoctoral award in Stellenbosch University

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Toolkits/Libraries —

Signal Processing –

Audio · Speech · Sensor · Feature Engineering · Noise-Robust Analysis · ASR (Whisper, Wav2Vec2, HuBERT) · Speaker Diarization · VAD (Silerio) · Speech Enhancement · openSMILE · jiwer · Librosa · SoundFile · TorchAudio · PyDub · SoX · FFmpeg · Kaldi · OpenCV · PyFeat

Machine Learning –

Deep Learning · Supervised Learning · Classification & Regression · Multimodal AI (Audio + Video + Text) · Transfer Learning · Model Evaluation · Feature Selection · Explainable AI · Time-Series Analysis · Graph Neural Networks · XGBoost · LightGBM · PyTorch · TensorFlow · Keras · Hugging Face · Scikit-learn

NLP & LLMs –

Prompt Engineering · Retrieval-Augmented Generation (RAG) · Embeddings & Semantic Search · LangChain · LangFuse · LangSmith · OpenAI · Groq · Jina AI · spaCy · NLTK · Gensim · SentenceTransformers

Healthcare & Clinical AI –

Clinical Decision Support · Digital Audio (cough, breath, speech) & Sensor Biomarkers · TB, COVID-19, Cognitive Decline, Mental Health Detection · AI Fairness & Ethics · Clinical Data

Software & Data Engineering –

End-to-End ML Pipelines · API Development (REST, FastAPI) · Asynchronous Programming · Real-Time Streaming (SSE) · Large-scale Data Engineering (ETL)

Cloud & DevOps –

Cloud Architecture (GCP) · Containerisation (Docker) · CI/CD Pipelines (GitHub Actions, Git CLI) · Serverless Computing · GCP · Azure AI Foundry · Docker · Vercel · Render · Linux/Unix

Research and Professional Experience

Jan 2024–

Research Fellow in Speech Processing and Machine Learning (intersection of speech, AI and healthcare, Grade 8) University of Sheffield, UK

➤ Research responsibilities

- Led development of multimodal machine learning systems for large-scale clinical speech and video analysis within the CognoSpeak, CognoMND, and CognoStroke programmes, supporting early detection of dementia, motor neuron disease, and stroke-related cognitive impairment.
- Designed and maintained end-to-end multimodal ML pipelines covering data ingestion, quality control, integrating acoustic speech features, ASR-derived linguistic representations, and facial behavioural analysis, and brought research ideas to production as an AI engineer [🔗 repo].
- Applied advanced ML methods for classification and regression modelling of heterogeneous clinical datasets.
- Managed and analysed longitudinal clinical datasets comprising 5,000+ NHS participants in collaboration with clinicians, incorporating cognitive and mental health assessments (MoCA, PHQ-9, GAD-7).
- Integrated and identified cognitive domain knowledge (attention, memory, and executive function) into data curation and model development to improve robustness and clinical interpretability.
- Developed interpretable biomarker-driven machine learning models characterising speech and behavioural markers associated with neurological and cognitive decline.
- Investigated algorithmic bias and fairness in clinical AI across multilingual and ethnically diverse UK populations.
- Disseminated research outcomes through peer-reviewed journal publications, conference presentations, and interdisciplinary collaborations.

➤ Team Lead and Supervision

- Led research activities contributing to peer-reviewed publications and public dataset releases PROCESS-2)
- Supervised COM4520 Darwin project group
- Supervised undergraduate, MSc, and PhD researchers working on pathological speech analysis and clinical AI.

➤ Research Project Management

- Coordinated research milestones and deliverables across interdisciplinary academic, clinical, and industry partners
- Provided technical leadership and data governance support within a multidisciplinary research environment

➤ Research Dissemination and Engagement

- Promoted CognoSpeak research through academic workshops and invited presentations at CMI, UKIS 2024, and MultiModalAI'24.
- Engaged public audiences through outreach activities including the BrightMinds Webinar and Pint of Science, communicating advances in AI-driven cognitive health assessment to the general public.

➤ Interdisciplinary Collaboration and Engagement

- as an event organiser: Organise interdisciplinary workshops promoting AI applications in mental health and digital healthcare
- with academic partners: SITraN (Neurology), Prof. Annalena Venneri (Psychology), Dr. Chaona Chen (Robotics), Prof. Pamela Enderby (Speech and Language Therapy)
- with industry partners: TherapyBox

➤ Decision making and academic support

- Member of the Senate Appeals and Discipline Panels at the University of Sheffield (effective date: 2024 to 2027)

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Toolkits/Libraries —

Frontend & UI –
Interactive UI Development · Web Frameworks (Next.js, React) · Tailwind CSS · JavaScript · TypeScript · HTML5 · CSS3

Programming –
Python · Anaconda · SQL · Bash · PowerShell · MATLAB · R · Java

Databases –
FAISS · ChromaDB · Pinecone · PostgreSQL · MySQL · Firebase · Cloud Firestore

Backend/APIs –
FastAPI · Django · Flask · REST APIs · Redis · Microservices

Data Science/Engineering –
Pandas · NumPy · SciPy · Matplotlib · Seaborn · Ptitprince · Plotly · Openpyxl · Tabulate · n8n

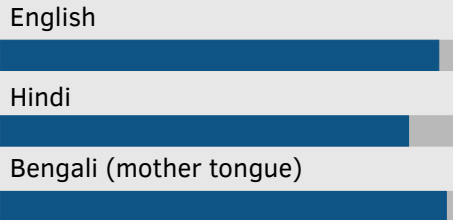
MLOps & Experimentation –
MLflow · Weights & Biases (wandb) · TensorBoard · Optuna · Hyperparameter Tuning

GPU & Parallel Computing –
CUDA · Multi-GPU Training (PyTorch) · Distributed Data Parallel (DDP) · Mixed Precision Training (AMP) · cuDNN · GPU Optimisation · HPC · Multiprocessing · ThreadPool

Bioinformatics –
H5Py · PySAM · Tombo · Guppy · Minimap2 · SAMtools · LongReadSum · ont_fast5_api

AI Coding Assistants –
Claude · Codex

Languages Known —



Research and Professional Experience, cont.

Feb 2023–
Dec 2023

Research Scholar

University of Nevada, Las Vegas (UNLV), USA

- Developed deep learning methodologies for the detection of RNA modifications including 5-methylcytosine (5mC) and N6-methyladenine (m6A), using Oxford Nanopore long-read sequencing data.
- Designed machine learning frameworks for variant detection and transcriptomic analysis from RNA sequencing data.
- Investigated isoform-level genomic variation, including gene fusions and alternative splicing, using deep learning approaches.
- Processed and analysed large-scale ONT sequencing data (human and yeast) to support model training and benchmarking.
- Developed bioinformatics pipelines converting raw FAST5 signals into FASTQ/BAM representations for downstream analysis.
- Applied computational modelling towards biomarker discovery in neuropsychiatric disorders, including schizophrenia.
- Collaborated with multidisciplinary genomics researchers and supervised graduate and undergraduate students contributing to computational genomics projects.

Aug 2019–
Jan 2023

Post Doctoral Fellow

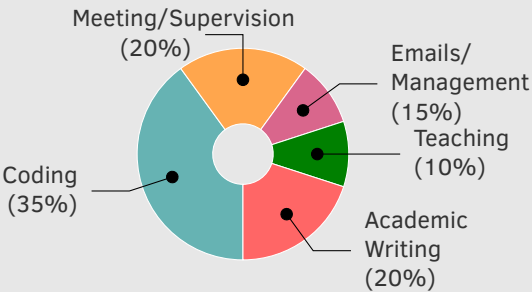
Stellenbosch University, South Africa

- Developed machine learning systems for detection of tuberculosis (TB) and COVID-19 from cough, breath, and speech recordings collected in real-world clinical environments
- Designed deep neural network architectures incorporating autoencoders, attention mechanisms, transfer learning, and generative models for respiratory disease classification
- Designed signal-processing workflows for noise-robust analysis of real-world speech, cough, and respiratory audio, evaluating how environmental noise, recording-device variability, and signal quality influence model performance
- Modelled cough frequency and recovery trajectories using acoustic analysis, language-model-based temporal modelling
- Built AI-driven patient monitoring solutions integrating audio sensing and accelerometer data through custom Android applications in real-world environments
- Conducted longitudinal monitoring of patient recovery and clinic activity through multimodal sensor analysis
- Managed large-scale biomedical audio datasets and optimised deep learning workflows using parallel computing
- Participated in international biomedical imaging challenges applying machine learning models for disease classification from chest X-ray images
- Investigated transfer learning using pre-trained speech representations (x-vectors, i-vectors, and bottleneck features) for cougher identification in under-resourced languages under noisy and real-world recording conditions
- Senior Investigator in the international (CAGE-TB) study
- Architected and deployed an end-to-end clinical data acquisition and processing pipeline, integrating microphones, servers, and mobile sensing platforms, and developing scalable Python workflows for preprocessing, annotation management, storage, and model-ready dataset generation
- Developed standard operating procedures (SOP) and trained healthcare workers in the deployment of AI-enabled sensing technologies within TB clinics
- Co-supervised honours and MSc students working on pathological speech and cough analysis using reinforcement learning, transformers, and NLP methods
- Delivered deep learning training materials covering CNN, LSTM, and ResNet architectures using PyTorch and TensorFlow

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A Typical Day



Other Projects

Enterprise RAG Platform: Modular AI Assistant System ([🔗](#) [repo](#) [🔗](#));
Territory Map Management | Full-Stack Firebase Web Application | Cloud Geospatial Data Engineering System (Live Demo | [🔗](#) [repo](#) [🔗](#))

In the Media

What your speech could reveal about your brain health;
Advertising our COVID research in the media (news, YouTube);

Certificates/Skills

First Aid for Mental Health, Sheffield, UK;
Action Counters Terrorism Awareness; Trauma-informed Training for Panel members;
Good Clinical Practice certificate;
QA Level 2 Award in Emergency First Aid at Work by Highland First Aid Training, Ross-shire, UK;
Level 2 NVQ Certificate in Spectator Safety (UK);
A passionate interest in Music;
Red Cross Social worker;
Oxfam festival volunteer;

Hobbies

Level 8 Local Guide (Google);
Latin American ballroom dancer;
Violin player (beginner level);
Les Mills Bodypump;
Mountain hiker
Active in sports (badminton, football)

Research and Innovation Projects

Jan 2024–

Automatic early detection of cognitive and mental health disorders from longitudinal conversational multimodal data (speech and video)
University of Sheffield, UK

- Develop scalable AI systems to enable early identification of dementia, mild cognitive impairment, depression, anxiety, motor neuron disease, and post-stroke cognitive decline (CognoS-peak, CognoMND and CognoStroke)
- Establish clinically interpretable speech and behavioural biomarkers supporting objective cognitive assessment and clinical decision-making
- Curate and quality-control longitudinal multimodal datasets from 5,000+ NHS participants in collaboration with clinicians to enable real-world clinical AI research
- Investigate relationships between attention, memory, and executive function measures and speech-derived behavioural markers of cognitive decline

2021–2024

Cough Audio triAGE for TB CAGE-TB Stellenbosch University, South Africa

- Develop non-invasive, low-cost AI screening approaches for tuberculosis and respiratory diseases using cough audio collected in real clinical environments across South Africa and Uganda
- Build large-scale cough datasets supporting robust machine learning model development for global health diagnostics
- Contribute to public release of rare clinical audio datasets to accelerate research in AI-based respiratory disease detection at the end of the CAGE-TB project

2023

COBRE, NIH project

UNLV, USA

- Modification detection: Study the relationship of base modifications to disease pathology at the genomic/transcriptomic scale using Oxford Nanopore sequencing and deep learning
- Long-read data analysis: Develop computational tools via machine learning especially deep learning for genomic and transcriptomic data analysis on long-read sequencing data

2020–2023

Detecting COVID-19 in vocal audio

Stellenbosch University, South Africa

- Develop machine learning systems analysing cough, breath, and speech signals for rapid remote COVID-19 detection
- Model disease severity and hospitalisation risk using large-scale international clinical audio datasets

2019–2023

Long-term patient monitoring

Stellenbosch University, South Africa

- Automatic detection of coughs among other environmental noises in a multi-bed ward at TASK (a 24h TB research centre)
- Enable automated cough detection and bed occupancy monitoring in hospital wards to support longitudinal healthcare analytics using a non-invasive smartphone-inbuilt accelerometer

2019–2021

TB cough classification

Stellenbosch University, South Africa

- Develop robust machine learning models distinguishing TB coughs from other respiratory conditions under noisy real-world recording conditions

2010–2014

A MEMS/CMOS microphone

University of Edinburgh & Stirling, UK

- Design and develop a multi-channel adaptive MEMS/CMOS microphone system inspired by human auditory processing

References

Available on request